

ME 300 Homework 9: Isentropic Efficiency and Cycle Analysis

Answer the following questions neatly on a separate piece of paper or on a tablet. Written answers can be typed. Show all your work, list all your assumptions, and explain your reasoning – partial credit will only be given if you show and explain each of your steps. Submit your homework on Canvas.

1. (25 points) Consider the following steam turbine with a mass flow rate of 15 kg/s.
 - a. Measurements from the power plant of the actual (non-ideal) turbine show that the inlet pressure is $P_1=12$ MPa and inlet temperature is $T_1=640$ K. The outlet pressure is $P_2=1$ MPa and a quality of $x_2=0.86$. Calculate the work output of this machine and the change in the entropy of the fluid across the turbine.
 - b. Recalculate the work for an ideal turbine assuming the same initial state and final pressure, and use that to calculate the component efficiency.
 - c. Would this process be possible if the final quality is 0.8? Why or why not?
2. (15 points) Use the data provided from the West Campus Steam Plant at the University Park campus – the data shows the pressure and temperature at the inlet (P_1, T_1) and outlet (P_2, T_2) of the compressor of the gas turbine. Use the data to calculate the isentropic efficiency of the compressor at each hour of the data provided and plot the data vs. hour. Identify any assumptions you made and write out your calculation process before implementing it in Matlab or Excel. Discuss your final plot and what's happening to the isentropic efficiency with time. What sorts of things could the power plant operators do to increase the isentropic efficiency?
3. (10 points) Explain the difference between a heat pump and a refrigerator. Use their performance metrics to help support your argument.
4. (15 points) You find specifications for a refrigerator that say the following:
 - Power requirement: 1000 W
 - $T_L = 0^\circ\text{C}$
 - $T_H = 20^\circ\text{C}$
 - a. Draw a diagram for this refrigerator showing the heat and work flows as well as the two temperature reservoirs.
 - b. What is the maximum coefficient of performance that you could achieve with this refrigerator?
 - c. You run some tests and find that the coefficient of performance is actually 7. Determine the heat in and heat rejected from this refrigerator.
5. (35 points) Consider the following Rankine Cycle power plant. Saturated water vapor at 8 MPa enters the turbine of a 100 MW (net power) power plant. The pressure at the exit of the turbine is 0.008 MPa. Saturated liquid leaves the condenser at 0.008 MPa. The turbine and pump have isentropic efficiencies of 85%. Calculate the mass flow rate of water through the cycle as well as the power generated by the turbine component. Write all your assumptions.